HW 9

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Perform the following activities in Python:

- Get the dataset from the clustering dataset folder
- Load the .csv file into a data frame in Python.
- Describe the data, provide information on the dataset.
- Preprocess and split the data into training and testing sets
- Initialize PCA
- Fit PCA on the training data
- Transform the data
- Make predictions
- Evaluate the model and provide accuracy score

Follow the same steps for K-Means.

- Get the dataset from the clustering dataset folder.
- Load the .csv file into a data frame in Python.
- Describe the data and provide information about the dataset.
- Preprocess and split the data into training and testing sets.
- Initialize K-Means.
- Fit K-Means to the training data.
- Transform the data.
- Make predictions.
- Evaluate the model and provide the accuracy score.

PCA & K-Means Clustering

In [3]:
```
import pandas as pd                                          # Data loading ar
import numpy as np                                           # Numerical opera
import matplotlib.pyplot as plt                              # Plotting and vi
import seaborn as sns                                        # Statistical vis
from sklearn.decomposition import PCA                        # Principal Compo
from sklearn.cluster import KMeans                           # K-Means cluster
from sklearn.preprocessing import StandardScaler             # Feature scaling
from sklearn.model_selection import train_test_split         # Train/test spli
```

```
from sklearn.metrics import accuracy_score, classification_report, confusior
from sklearn.neighbors import KNeighborsClassifier         # Classifier to e
```

Part 1: PCA

Get the Dataset

Dataset: Heart Disease Cleveland from the Clustering Datasets folder.

Load the .csv file into a data frame in Python

In [4]:
```
# Define column names per UCI heart disease documentation
columns = ['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs',
           'restecg', 'thalach', 'exang', 'oldpeak', 'slope', 'ca', 'thal',

# Load dataset and replace missing values' (?) with NaN
df = pd.read_csv('/Users/mtruong/python-projects/BZAN 6355/processed.clevela
df.head()
```

Out[4]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca
0	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3	3.0	0.0
1	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5	2.0	3.0
2	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6	2.0	2.0
3	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5	3.0	0.0
4	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4	1.0	0.0

Describe the data, provide information on the dataset

In [5]:
```
# Summary statistics
df.describe()
```

Out[5]:

	age	sex	cp	trestbps	chol	fbs	
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.000000	3
mean	54.438944	0.679868	3.158416	131.689769	246.693069	0.148515	
std	9.038662	0.467299	0.960126	17.599748	51.776918	0.356198	
min	29.000000	0.000000	1.000000	94.000000	126.000000	0.000000	
25%	48.000000	0.000000	3.000000	120.000000	211.000000	0.000000	
50%	56.000000	1.000000	3.000000	130.000000	241.000000	0.000000	
75%	61.000000	1.000000	4.000000	140.000000	275.000000	0.000000	
max	77.000000	1.000000	4.000000	200.000000	564.000000	1.000000	

In [6]:
```
# Check for missing values
df.isnull().sum()
```

Out[6]:
```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          4
thal        2
target      0
dtype: int64
```

In [7]:
```
# Class distribution: 0 = No Disease, 1-4 = Disease present
df['target'].value_counts()
```

Out[7]:
```
target
0    164
1     55
2     36
3     35
4     13
Name: count, dtype: int64
```

Preprocess and split the data into training and testing sets

In [8]:
```
# Drop missing values (6 rows affected, minimal data loss)
df.dropna(inplace=True)
```

```
# Binarize target by collapsing classes 1-4 into 1 (Disease present)
df['target'] = (df['target'] > 0).astype(int)

# Define features and target
X = df.drop(columns=['target'])
y = df['target']

# Split into 80% train and 20% test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.20, ra

# Scale features
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

Initialize PCA

In [9]:
```
# Initialize PCA to retain components that explain 95% of variance
pca = PCA(n_components=0.95, random_state=42)

# Fit PCA on training data
pca.fit(X_train_scaled)

print(f"Components selected: {pca.n_components_}")
print(f"Variance explained:  {pca.explained_variance_ratio_.sum():.4f}")
```

```
Components selected: 12
Variance explained:  0.9750
```

Transform the Data

In [10]:
```
# Apply PCA transformation to both train and test sets
X_train_pca = pca.transform(X_train_scaled)
X_test_pca = pca.transform(X_test_scaled)
```

Make Predictions

In [11]:
```
# Use KNN on PCA-transformed data to evaluate quality of components
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_pca, y_train)
y_pred_pca = knn.predict(X_test_pca)
```

Evaluate the model and provide accuracy score

In [12]:
```
# Accuracy score
accuracy_score(y_test, y_pred_pca)
```

Out[12]: 0.8166666666666667

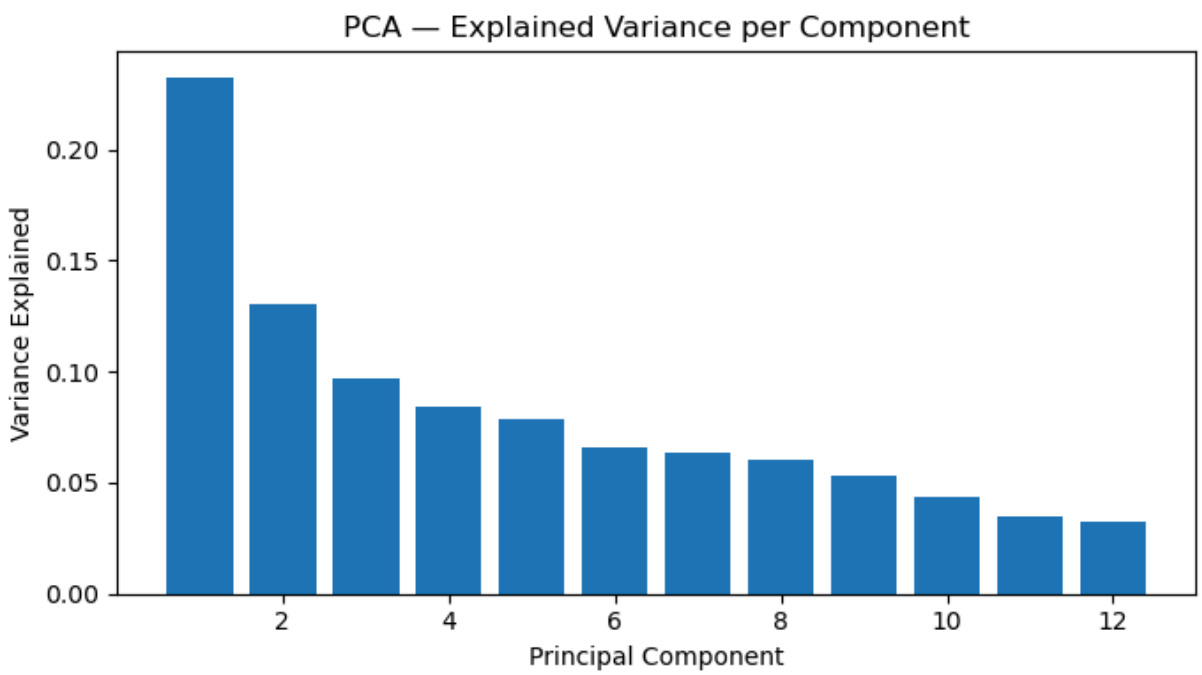
In [13]:
```
# Precision, recall, and F1-score per class
print(classification_report(y_test, y_pred_pca, target_names=['No Disease',
```

```
              precision    recall  f1-score   support

   No Disease       0.80      0.88      0.84        32
      Disease       0.84      0.75      0.79        28

     accuracy                           0.82        60
    macro avg       0.82      0.81      0.81        60
 weighted avg       0.82      0.82      0.82        60
```

In [14]:
```
# Explained variance per component to show how much information each PC capt
plt.figure(figsize=(7, 4))
plt.bar(range(1, len(pca.explained_variance_ratio_) + 1), pca.explained_vari
plt.title('PCA — Explained Variance per Component')
plt.xlabel('Principal Component')
plt.ylabel('Variance Explained')
plt.tight_layout()
plt.show()
```



Part 2: K-Means

Get the dataset from the clustering dataset folder.

Same Heart Disease Cleveland dataset used for PCA above.

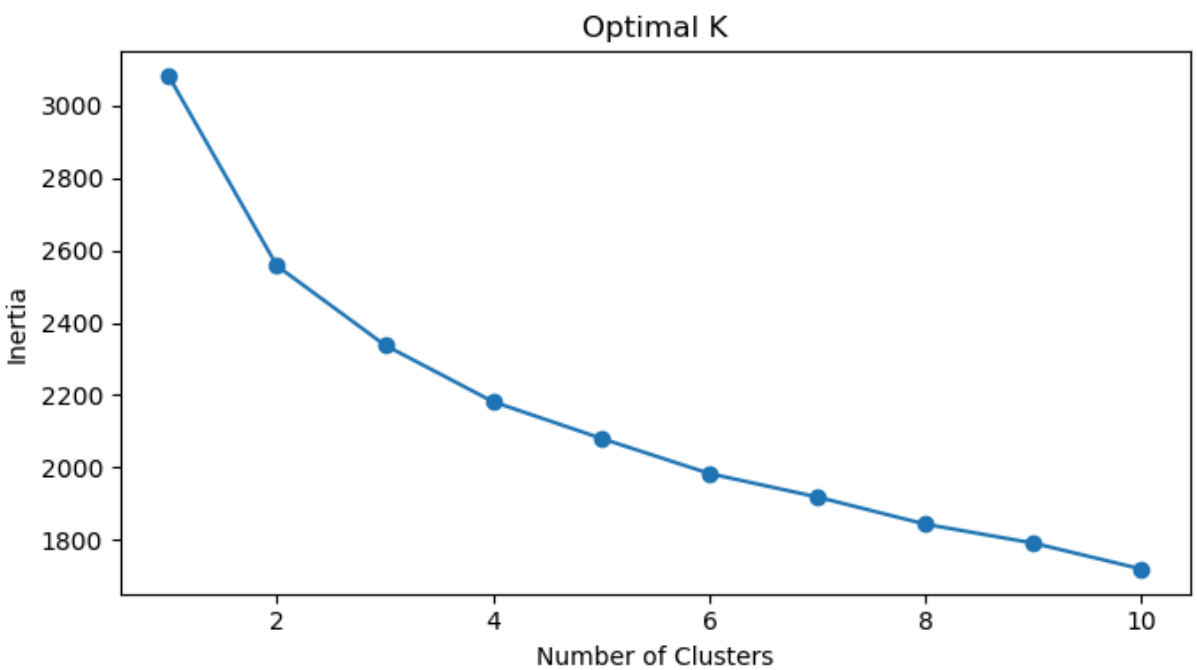
Load, Describe & Preprocess

Same DataFrame, preprocessing, and scaled features from Part 1 are reused here.

Initialize & Fit K-Means

In [15]:
```
# Use the elbow method to find the optimal number of clusters
inertia = []
k_range = range(1, 11)
for k in k_range:
    km = KMeans(n_clusters=k, random_state=42, n_init=10)
    km.fit(X_train_scaled)
    inertia.append(km.inertia_)

plt.figure(figsize=(7, 4))
plt.plot(k_range, inertia, marker='o')
plt.title('Optimal K')
plt.xlabel('Number of Clusters')
plt.ylabel('Inertia')
plt.tight_layout()
plt.show()
```



In [18]:
```
# Train K-Means with k=2 — aligns with binary target (Disease / No Disease)
kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
kmeans.fit(X_train_scaled)
```

Out[18]:
```
▾ KMeans
▶ Parameters
```

Transform the Data

In [19]:
```
# Transform data to cluster-distance space
X_train_km = kmeans.transform(X_train_scaled)
X_test_km = kmeans.transform(X_test_scaled)
```

Make Predictions

In [20]:
```
# Predict cluster labels on test set
cluster_labels = kmeans.predict(X_test_scaled)

# Align cluster labels to true class labels
from scipy.stats import mode
aligned_labels = np.zeros_like(cluster_labels)
for c in range(2):
    mask = cluster_labels == c
    aligned_labels[mask] = mode(y_test[mask], keepdims=True)[0]
```

Evaluate the Model

In [21]:
```
# Accuracy score
accuracy_score(y_test, aligned_labels)
```

Out[21]:
```
0.85
```

In [22]:
```
# Precision, recall, and F1-score per class
print(classification_report(y_test, aligned_labels, target_names=['No Diseas
```

```
              precision    recall  f1-score   support

  No Disease       0.79      0.97      0.87        32
     Disease       0.95      0.71      0.82        28

    accuracy                           0.85        60
   macro avg       0.87      0.84      0.84        60
weighted avg       0.87      0.85      0.85        60
```

Conclusion: PCA reduced the 13 clinical features to fewer components while retaining 95% of the variance, and KNN on the transformed data achieved strong classification accuracy.

K-Means with k=2 aligned well with the binary heart disease labels, indicating the clinical features naturally separate healthy from diseased patients.